Gene expression analysis of influenza A virus-infected cells to identify biomarkers using Random Forests

Report of the bioinformatics practical course, Project 5

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Abstract

Influenza A virus (IAV) infection triggers the loss of SUMO-modified TRIM28, a transcriptional co-repressor of endogenous retroviruses, and this switch amplifies the interferon (IFN) response of the infected cell. The RNA-seq data of that study (GEO GSE133329: A549 cells expressing wild-type (wt) TRIM28 or the SUMOylation-deficient TRIM28 6KR mutant, mock or IAV infected, three replicates each) were reanalysed to find genes that separate the four conditions and could serve as biomarker candidates. Differential expression analysis with DESeq2 found 8,545 genes changed by infection in wt cells and 95 genes whose infection response differs between the genotypes; those with a higher infection response in 6KR cells are enriched for antiviral defence and type I IFN signalling. A Random Forest classifier was evaluated by nested leave-one-out cross-validation with the gene selection repeated inside every fold. The planned model, which selects genes by their overall difference between the four groups, classified 8 of 12 samples correctly. All errors were genotype errors, most likely because infection genes dominated the selection. Selecting genotype and infection genes separately, a rule derived from this error analysis, classified 12 of 12 samples correctly (95 % confidence interval 74–100 %); this estimate is optimistic, because the rule was chosen after the first result. The candidate genes of this model are mostly genotype markers, among them five KRAB zinc-finger genes, known readouts of TRIM28 repressor function, while the apparent induction of the infection marker RPL30 most likely reflects relative resistance to the virus-induced host shutoff. With twelve samples from one cell line, the candidates mainly describe the cellular TRIM28 state and need independent validation.

1. Introduction

Influenza A virus (IAV) infects the epithelium of the respiratory tract. Infected epithelial cells sense viral RNA through cytosolic receptors such as RIG-I and respond with type I and type III interferons (IFNs), which induce hundreds of IFN-stimulated genes (ISGs) with antiviral function. IAV counteracts this response with its non-structural protein 1 (NS1), which binds double-stranded RNA (dsRNA) and blocks the activation of the transcription factor IRF3 and thereby the induction of IFN-β[1].

The tripartite motif protein TRIM28 (also KAP1) is the co-repressor of the Krüppel-associated box zinc-finger proteins (KRAB-ZNFs), the largest family of transcriptional regulators in vertebrates, which silence transposable elements such as endogenous retroviruses (ERVs)[2]. In adult human cells, loss of TRIM28 derepresses ERVs and ZNF genes and triggers an RNA-sensing response through the adaptor MAVS[3]. Schmidt et al. showed that IAV infection causes the loss of SUMO-modified TRIM28, which derepresses ERVs and unmasks them as a cellular source of dsRNA that amplifies the RIG-I/MAVS-dependent IFN response[4]. A TRIM28 mutant in which the six SUMOylated lysines are replaced by

arginines (TRIM28 6KR) mimics this state. Other viruses appear to target the same switch, which supports the idea of a TRIM28-controlled antiviral defence[5]. The RNA-seq data of that study, A549 cells reconstituted with wild-type (wt) TRIM28 or TRIM28 6KR, mock or IAV infected, are the data of this project.

Host transcriptional signatures can detect influenza infection, in blood even before the peak of symptoms[6–8]. Random Forests[9] are well suited to find such genes in expression data: they remain usable when there are thousands of genes but only a few samples, a setting in which classical regression models cannot be fitted without strong constraints, they classify more than two classes directly, and they rank genes by their importance for the classification[10]. With few samples, however, the performance estimate of a classifier is only trustworthy if every data-driven step, including the gene selection, is repeated inside the cross-validation[11,12].

The aim of this project was to analyse the GSE133329 count data with DESeq2 and a Random Forest classifier in R, to identify genes that separate the four conditions (wt or 6KR, mock or infected) as potential biomarkers, to assess the classifier by accuracy, sensitivity and specificity, and to interpret the selected genes in the context of the host antiviral response. The analysis started during the practical course and was rebuilt afterwards as a reproducible pipeline; where the rebuilt analysis differs from the course analysis, this is stated in the Discussion.

2. Materials and methods

2.1 Data

Per-gene read counts of GEO series GSE133329[4] were used as deposited (featureCounts, Ensembl GRCh38 release 91). A CRISPR/Cas9 TRIM28 knockout clone of A549 lung epithelial carcinoma cells was reconstituted with wt TRIM28 or TRIM28 6KR by lentiviral transduction, and cells were mock infected or infected with IAV A/WSN/33 at a multiplicity of infection (MOI) of 10 for 6 hours (three independent replicates per group, 12 samples, mRNA sequencing). The 12 GEO count tables contain 21,386 genes in identical order; 3,931 of them have zero counts in all samples. They are identical to the count table used during the course.

2.2 Preprocessing and quality control

Genes with at least 10 counts in at least 3 samples (the size of the smallest group) were kept. Counts were transformed with the regularised logarithm (rlog) of DESeq2[13] in blind mode, so that no group information enters the transformation. Sample structure was examined by principal component analysis (PCA) of the 500 most variable genes, by Euclidean distances between samples on all filtered genes and by hierarchical clustering (average linkage).

2.3 Differential expression

DESeq2[13] was fitted with the design `~ genotype + infection + genotype:infection` and the reference levels wt and mock. From this single fit, five contrasts were extracted: the infection effect in wt and in 6KR cells, the genotype effect (6KR vs wt) in mock and in infected cells, and the interaction, which measures how the 6KR mutation changes the infection effect. Interaction-up genes have a higher infection effect in 6KR than in wt cells (stronger induction or weaker repression), interaction-down genes a lower one. P-values were adjusted by the Benjamini-Hochberg (BH)

method, and genes with an adjusted p-value below 0.05 were called differentially expressed (DEGs). Fold changes for plots and ranking were shrunk with ashr[14], which pulls fold changes towards zero in proportion to their uncertainty, so that large but noisy fold changes of weakly expressed genes do not dominate the ranking. Ensembl identifiers were mapped to gene symbols with org.Hs.eg.db; 203 of 13,483 tested genes have no symbol and are kept under their Ensembl identifier, and 112 map to several symbols, of which the first was kept.

2.4 Over-representation analysis

Up- and down-regulated DEGs were tested for over-represented terms of GO Biological Process, KEGG and Reactome with g:Profiler[15] (version e114_eg62_p19_27110d83), with all tested genes as background and BH-adjusted p-values below 0.05. For the two large contrasts (infection in wt, genotype in mock), only DEGs with an absolute shrunken log2 fold change above 1 were used; the interaction list was used without a fold-change cutoff. Terms with fewer than 10 or more than 1,000 genes were removed.

2.5 Random Forest classification

The four groups were classified with Random Forests (randomForest[16], 1,000 trees, default number of candidate genes per split) on rlog values. Performance was estimated by leave-one-out cross-validation (LOOCV): each of the 12 samples was predicted by a model built on the other 11. **Gene selection was repeated inside every fold on the 11 training samples only**, so the left-out sample never influenced which genes the model saw[11]. The gene filter and the blind rlog transformation use no group labels and were computed once. Two selection rules were compared, each with 50 genes:

- *Model 1 (planned):* the 50 genes with the smallest p-value in a DESeq2 likelihood-ratio test of the four-group design against an intercept-only model.
- *Model 2 (post-hoc):* the 25 genes with the smallest p-value for the genotype effect and the 25 genes with the smallest p-value for the infection effect of an additive DESeq2 model (~ `genotype + infection`). This rule was added after the errors of Model 1 were analysed and is the only variant that was tested.

From the 12 out-of-fold predictions, accuracy with an exact binomial 95 % confidence interval, a one-sided binomial test against the no-information rate of 0.25 (BH-adjusted over the two models), and per-class sensitivity and specificity were computed. The infected-vs-mock performance and the genotype accuracy were read off the same four-class predictions. As a descriptive check, not as tuning, both models were also run with 20 and 200 genes. A final model per rule was fitted on all 12 samples to rank genes by the scaled mean decrease in accuracy (randomForest default), and the number of folds in which each gene was selected was counted as its selection stability. Candidate biomarkers were defined in advance as genes of Model 2 that are among its 20 most important genes and were selected in at least 10 of 12 folds.

2.6 Software and reproducibility

The analysis ran in R 4.5.3 with DESeq2 1.50.2, ashr 2.2.63, gprofiler2 0.2.4, randomForest 4.7.1.2 and ggplot2 4.0.2. The pipeline (seven numbered scripts and `run_all.R`) reproduces every table and figure of this report from the GEO files; package versions are recorded in an `renv` lockfile, and all numbers in this report are read from the pipeline output when the report is built.

3. Results

3.1 Do the samples group as expected?

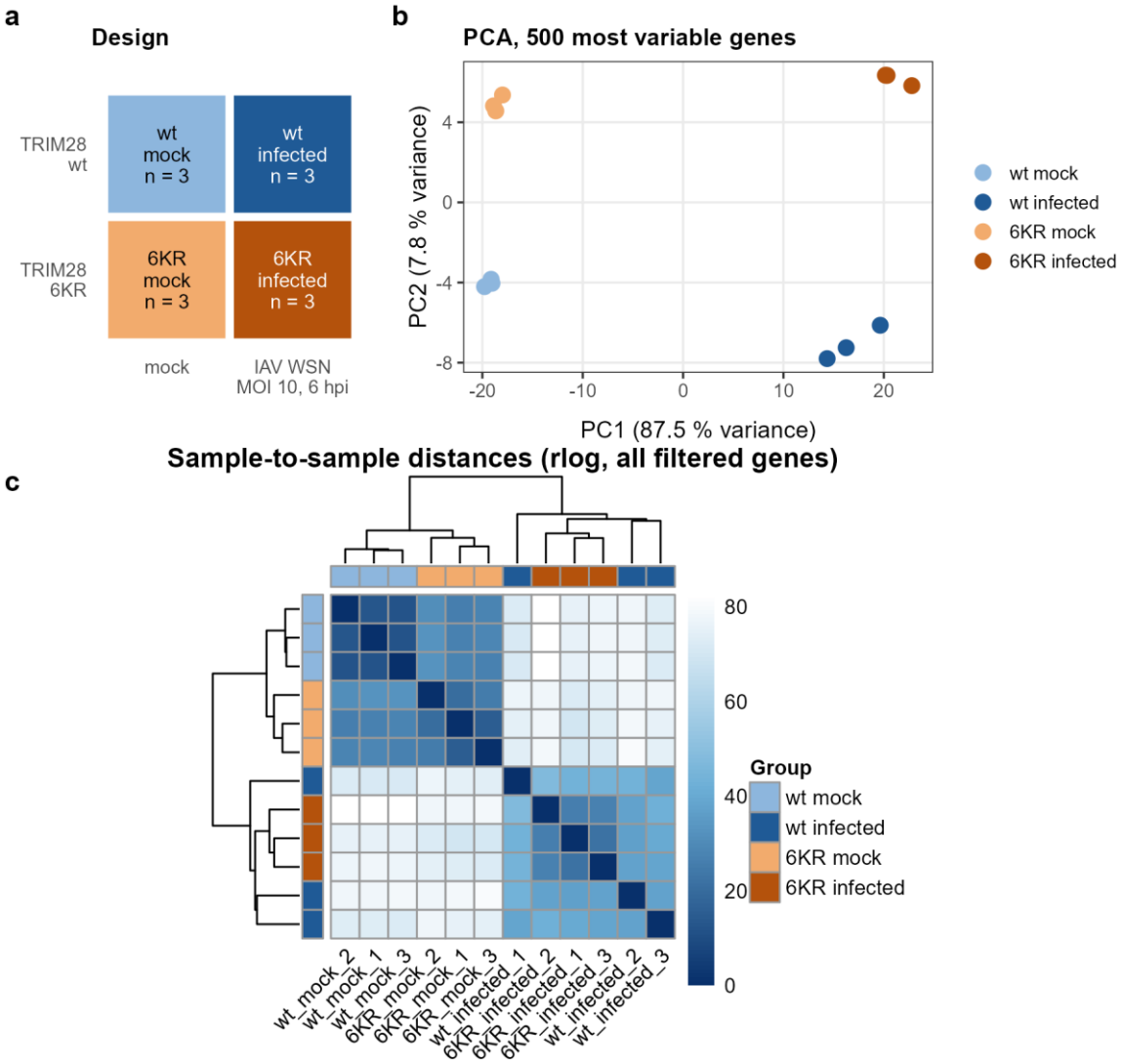


Fig. 1| **Study design and sample structure. a**, Design: a TRIM28 knockout clone of A549 cells reconstituted with wt TRIM28 or the SUMOylation-deficient TRIM28 6KR, mock or IAV A/WSN/33 infected (MOI 10, 6 hours post infection, hpi), three replicates per group. **b**, PCA of the 500 most variable genes after rlog transformation; the percentage of variance is given on each axis. **c**, Euclidean distances between samples on all 13,483 filtered genes, clustered by average linkage; darker blue means more similar.

The expression filter kept 13,483 of 21,386 genes. Library sizes ranged from 3.6 to 46.6 million assigned reads, and all six infected samples had smaller libraries than all six mock samples (Supplementary Fig. 1). In the PCA, the first principal component (87.5 % of the variance) separates infected from mock samples and the second (7.8 %) separates the genotypes (Fig. 1 b). The four groups form four separate clusters on the 500 most variable genes.

On all filtered genes, the wt-infected replicates are further apart from each other (mean distance to their replicates 37.7–40.7) than the replicates of the other groups (11.8–26.4), `wt_infected_2` lies closest to a 6KR-infected sample (Fig. 1 c), and in the dendrogram `wt_infected_1` joins the infected branch on its own (Supplementary Fig. 2). These three samples have the smallest libraries

(3.6–7.5 million reads) and the highest share of filtered genes with fewer than 5 counts (9.7–15.6 % against 1.6–6.8 %). The spread therefore most likely reflects noise in weakly expressed genes rather than a failed sample, and all 12 samples were kept.

3.2 Infection dominates, and the 6KR mutation amplifies the antiviral response

Infection changed the expression of 8,545 of 13,483 tested genes in wt cells (4,355 up, 4,190 down) and 8,754 genes in 6KR cells (Fig. 2 a). The 6KR mutation alone changed 1,475 genes in mock cells and 1,854 genes in infected cells. Classical ISGs were induced by infection in wt cells, for example OASL (shrunken log2 fold change 5.5) and IFIT1 (4.5) (Fig. 2 b), whereas IFNB1 itself was not significantly induced (adjusted p = 0.12).

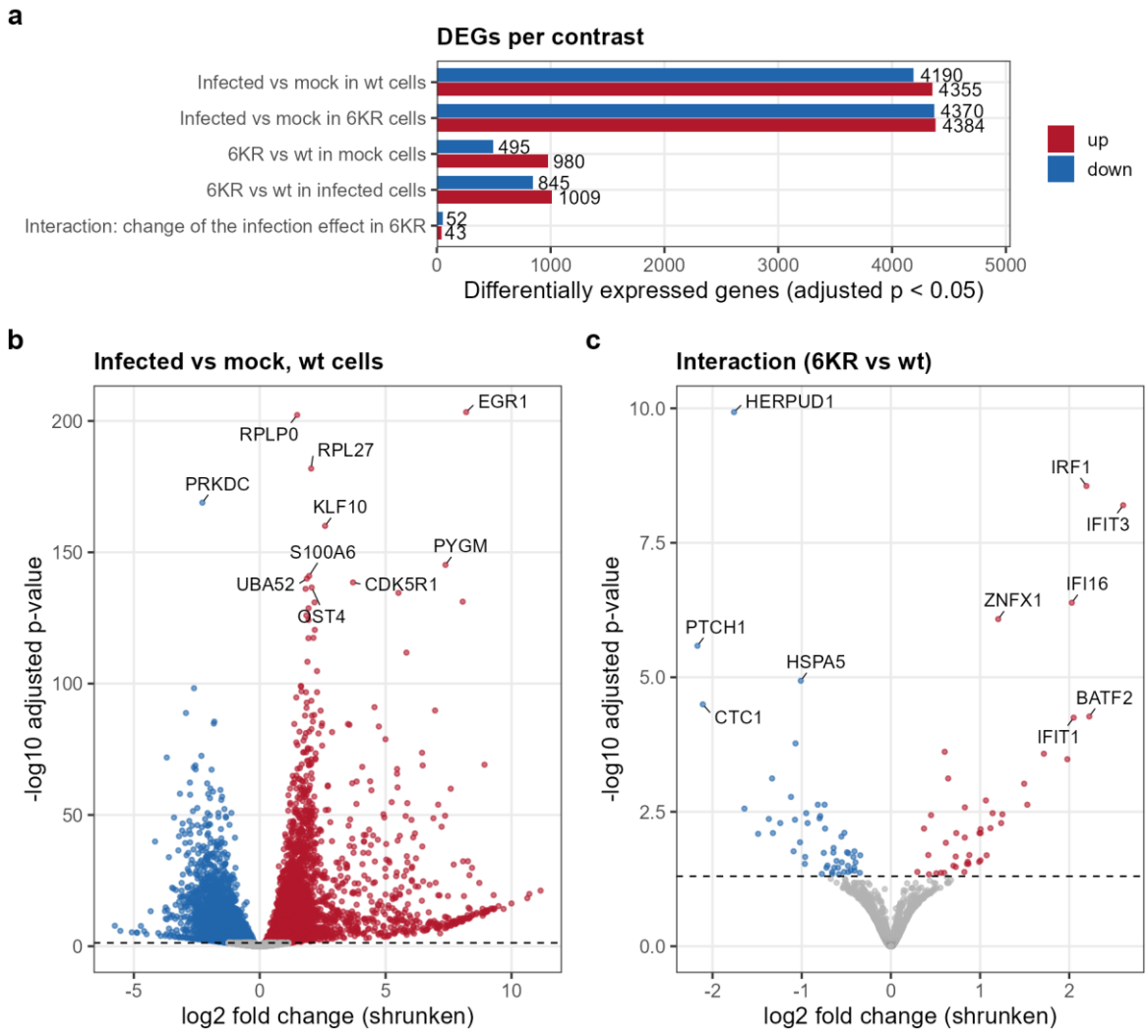


Fig. 2| **Differential expression. a**, Number of up- and down-regulated genes (BH-adjusted p < 0.05) per contrast of the DESeq2 interaction model. **b**, Infection effect in wt cells. **c**, Interaction: difference of the infection effect between 6KR and wt cells; positive values mean a higher infection effect in 6KR cells (stronger induction or weaker repression). Volcano plots show ashr-shrunken log2 fold changes; the dashed line marks an adjusted p-value of 0.05, and the ten most significant genes are annotated.

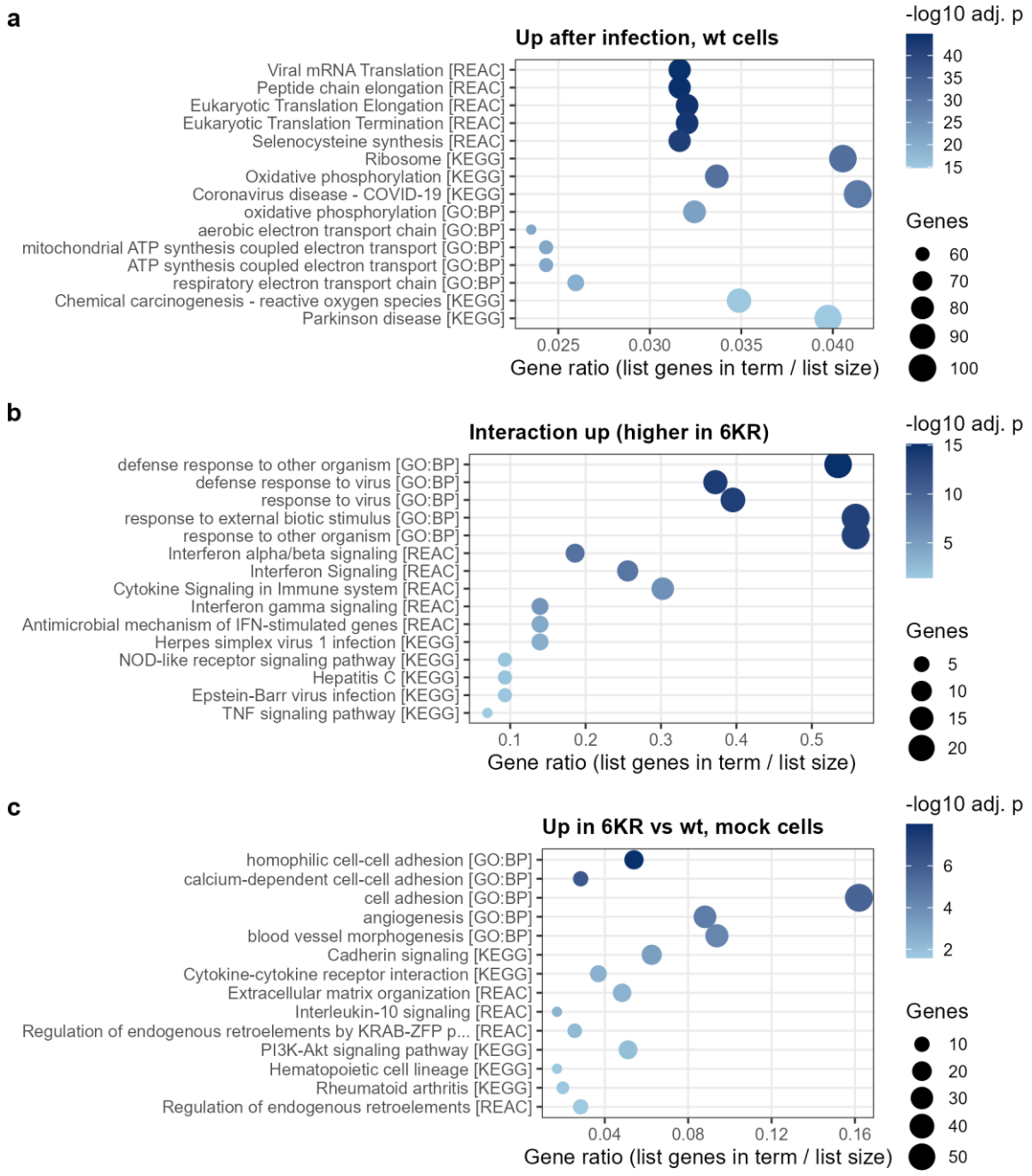


Fig. 3| **Over-representation analysis.** Five most significant terms per source (GO Biological Process, KEGG, Reactome) for **a**, genes induced by infection in wt cells; **b**, interaction-up genes (higher infection effect in 6KR than in wt cells); **c**, genes up-regulated in mock-infected 6KR vs wt cells. Colour gives the −log10 BH-adjusted p-value, point size the number of list genes in the term, and the x-axis the share of list genes in the term. Background: all 13,483 tested genes.

The interaction identified 95 genes whose infection response differs between the genotypes (43 up, 52 down; Fig. 2 c). Of the 43 interaction-up genes, 28 were induced by infection in 6KR cells (log2 fold change above 0), and 15 were repressed by infection in wt cells and less so in 6KR cells. The interaction-up genes include ISGs such as IRF1, IFIT3, IFI16 and USP18. IFNB1 itself was not a significant interaction gene (adjusted p = 0.20), but in infected cells it was 4.1 log2 units higher in 6KR than in wt cells (shrunken; adjusted p = 2.1×10^{-4}). Of 17 genes named as key IAV response genes of epithelial cells in a integrative analysis of 35 datasets[17], 7 are interaction-up genes (IFIT3, BATF2, USP18, HERC5, IFIT2, OAS1, OAS2). **The interaction-up genes therefore overlap with the canonical**

epithelial antiviral response. The interaction-down genes include ER stress genes such as HERPUD1 and HSPA5. The remaining contrasts are shown in Supplementary Fig. 3 and the 50 most significant interaction genes in Supplementary Fig. 4.

3.3 Which processes are affected?

The 2,474 genes induced by infection in wt cells gave 268 enriched terms, led by translation and ribosome terms and oxidative phosphorylation (Fig. 3 a). 71 of 77 cytosolic ribosomal protein genes and 55 of 77 nuclear-encoded oxidative phosphorylation subunit genes are in this list, while "Interferon alpha/beta signaling" only ranks at position 104. As discussed below, this pattern most likely reflects relative resistance of these transcripts to the virus-induced host shutoff. Genes repressed by infection were enriched for cell cycle, DNA replication and repair terms (Supplementary Fig. 5).

The 43 interaction-up genes gave 156 terms, led by defence response to other organisms and to viruses, followed by type I IFN production and signalling (adjusted $p = 1.1 \times 10^{-14}$ for "defense response to virus"; Fig. 3 b). The 352 genes up-regulated in uninfected 6KR cells were enriched for homophilic and calcium-dependent cell adhesion, and Reactome listed "Regulation of endogenous retroelements by KRAB-ZFP proteins" (adjusted $p = 0.0062$; Fig. 3 c), which reflects the many KRAB-ZNF genes in this list. The interaction-down genes were enriched for the unfolded protein response (Supplementary Fig. 5).

3.4 Random Forest classification: from the planned model to a design-aware gene selection

The planned Model 1 classified 8 of 12 samples correctly (accuracy 0.67, 95 % confidence interval 0.35–0.90, BH-adjusted $p = 0.0028$ against the no-information rate of 0.25; Table 1, Fig. 4 a). Infected and mock samples were separated without error (12 of 12), but all 4 errors confused the genotypes: wt_mock_3 was predicted as 6KR mock, wt_infected_2 was predicted as 6KR infected, 6KR_mock_2 was predicted as wt mock, 6KR_infected_2 was predicted as wt infected. The share of tree votes for the true class of these samples (0.38–0.46) was below that of the predicted class (0.51–0.61).

The most likely reason lies in the gene selection. The likelihood-ratio test ranks genes by their overall difference between the four groups, and this difference is dominated by infection: 50 of the 50 genes of the final Model 1 are infection DEGs, but only 7 are genotype DEGs in mock cells, with a median absolute genotype fold change of 0.05 log2 units. **The model saw almost no genotype information.** Model 1 also depended on the number of genes (9, 8 and 11 of 12 correct with 20, 50 and 200 genes; Fig. 4 c).

Model 2 applies the 2 × 2 design to the gene selection and takes half of its genes from the genotype effect and half from the infection effect. It classified 12 of 12 samples correctly (95 % confidence interval 0.74–1.00, BH-adjusted $p = 1.2 \times 10^{-7}$), including both genotypes in mock and infected cells (Fig. 4 b), and it did so with 20, 50 and 200 genes alike. Because the rule was derived from the errors of Model 1 on the same 12 samples, its cross-validated accuracy is an optimistic estimate; the pre-specified result of this project is the one of Model 1.

Table 1| **Random Forest performance in leave-one-out cross-validation.** Both models use 50 genes selected inside every fold. The infected-vs-mock and genotype results are read off the same four-class predictions. P-values are one-sided binomial tests against the no-information rate of 0.25, BH-adjusted over the two models. Sensitivity and specificity are computed per class (one class against the other three); a range is given where the four classes differ.

Measure	Model 1 (planned)	Model 2 (post-hoc)
Correct predictions (four classes)	8 / 12	12 / 12
Accuracy (95 % CI)	0.67 (0.35–0.90)	1.00 (0.74–1.00)
Adjusted p vs. chance	0.0028	1.2×10^{-7}
Sensitivity per class	0.67	1.00
Specificity per class	0.89	1.00
Infected vs. mock correct	12 / 12	12 / 12
Genotype correct, mock samples	4 / 6	6 / 6
Genotype correct, infected samples	4 / 6	6 / 6
Correct with 20 / 50 / 200 genes	9 / 8 / 11	12 / 12 / 12
Genes selected in all 12 folds	22	20

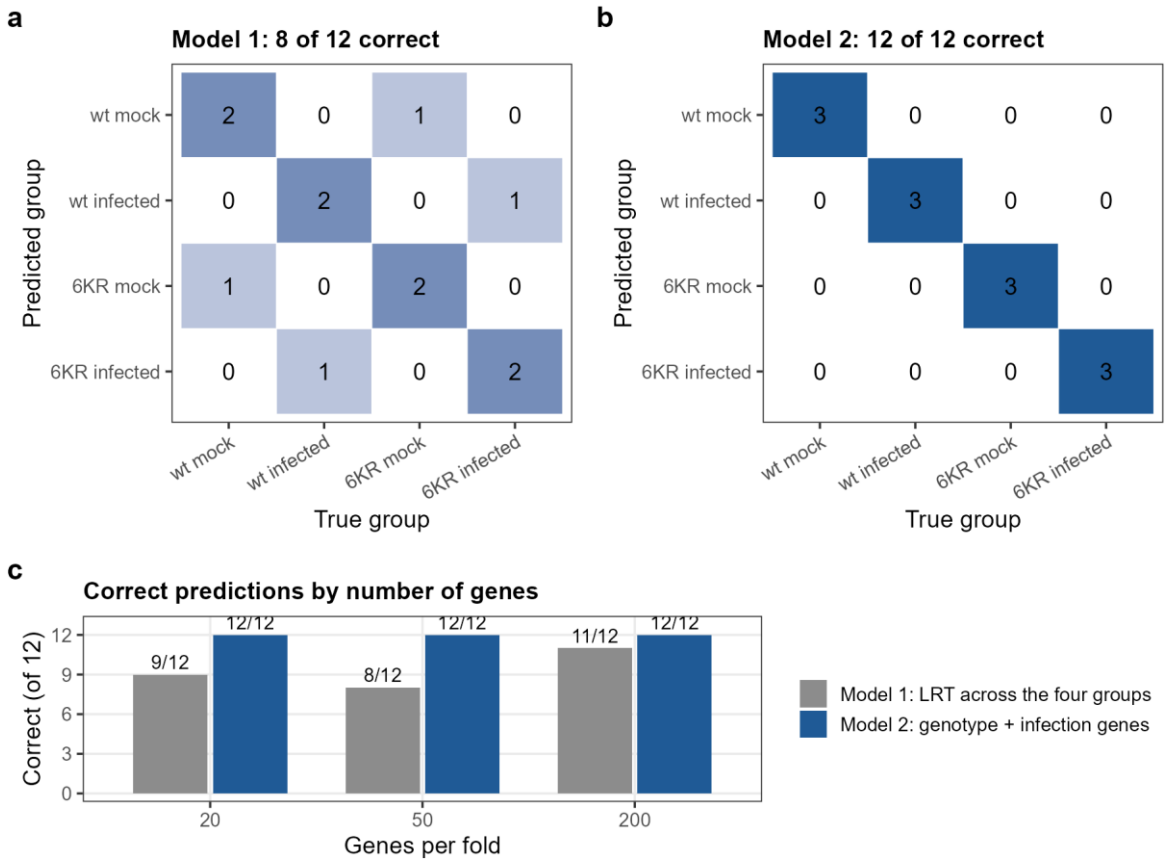


Fig. 4| **Random Forest classification. a**, **b**, Confusion matrices of the leave-one-out predictions of Model 1 (genes selected by a likelihood-ratio test across the four groups) and Model 2 (25 genotype and 25 infection genes). **c**, Number of correct predictions with 20, 50 and 200 genes per fold; this check was descriptive and not used to choose the gene number.

3.5 Candidate biomarkers

13 genes of Model 2 were among its 20 most important genes and were selected in at least 10 of 12 folds (Table 2, Fig. 5). 10 of them are genotype markers, 5 of these KRAB-ZNF genes, and ADAMTS2

shows the strongest genotype effect of all candidates (log2 fold change 7.5 in mock cells). The three infection markers are RPL30, ARRB2 and OST4. No classical ISG passed the stability rule: OASL, the most important gene of the final model, was selected in 9 of 12 folds.

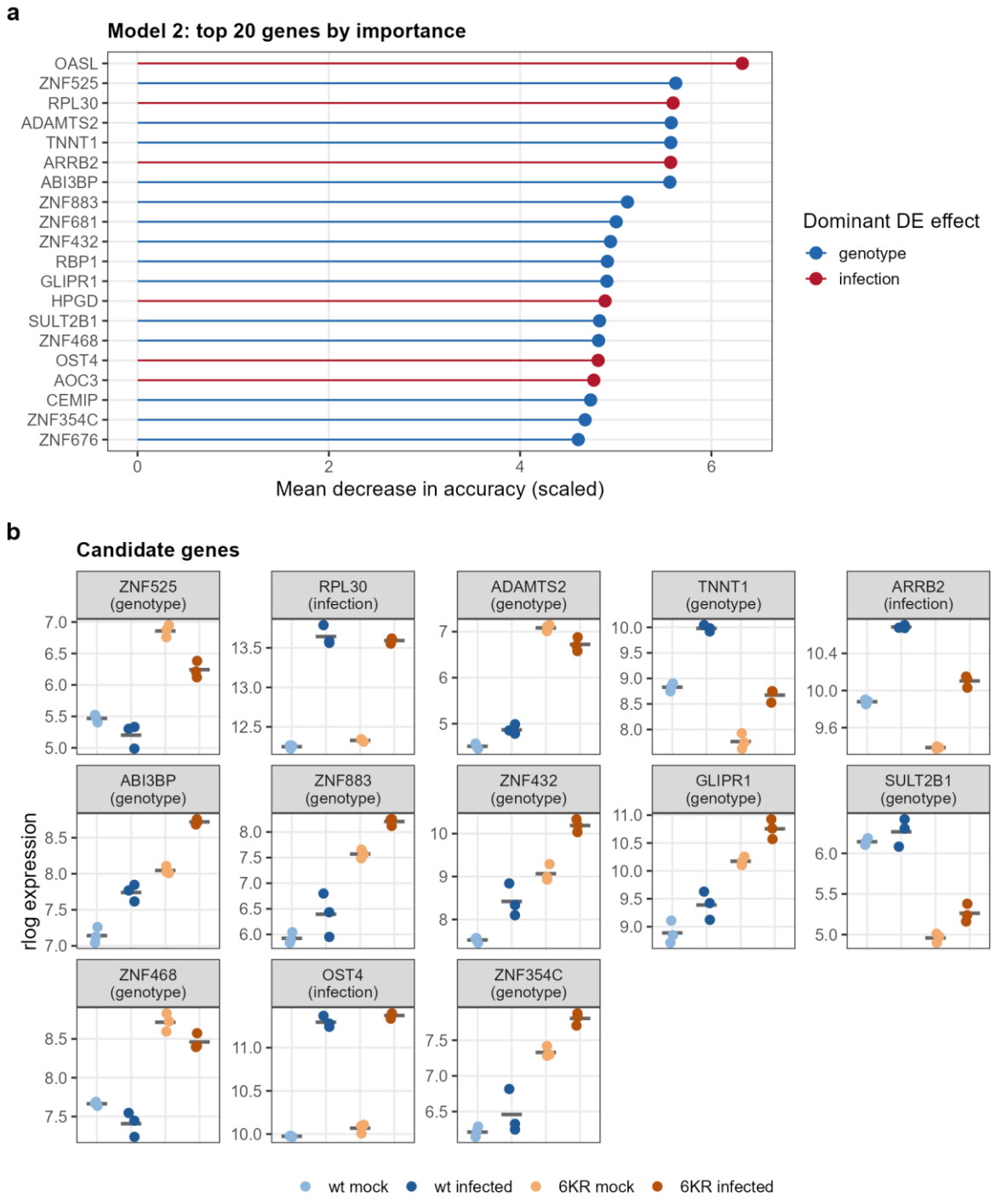


Fig. 5| **Candidate biomarkers. a**, The 20 most important genes of the final Model 2 (scaled mean decrease in accuracy), coloured by their dominant effect in the differential expression analysis (the factor with the larger significant shrunken fold change). **b**, rlog expression of the 13 candidate genes per sample; bars mark the group means.

Table 2| **Candidate biomarker genes.** Genes among the 20 most important genes of the final Model 2 that were selected in at least 10 of 12 cross-validation folds, ordered by importance. Log2 fold changes are ashr-shrunken: infection in wt cells and 6KR vs wt in mock cells. † Ribosomal protein gene; its induction is most likely relative (host shutoff, see Discussion).

Gene	Dominant effect	Importance rank	Folds selected	log2 FC infection (wt)	log2 FC 6KR vs wt (mock)
ZNF525	genotype	2	10 / 12	-1.27	2.83
RPL30 †	infection	3	10 / 12	2.13	0.08
ADAMTS2	genotype	4	12 / 12	0.37	7.46
TNNT1	genotype	5	12 / 12	1.68	-1.86
ARRB2	infection	6	12 / 12	1.21	-0.81
ABI3BP	genotype	7	12 / 12	1.11	1.62
ZNF883	genotype	8	12 / 12	1.14	3.26
ZNF432	genotype	10	12 / 12	1.70	2.71
GLIPR1	genotype	12	11 / 12	0.85	2.02
SULT2B1	genotype	14	10 / 12	0.28	-3.19
ZNF468	genotype	15	12 / 12	-0.49	1.72
OST4	infection	16	12 / 12	2.06	0.11
ZNF354C	genotype	19	12 / 12	0.57	2.16

4. Discussion

4.1 The 6KR mutation amplifies the interferon response to infection

The differential expression results reproduce the central transcriptomic finding of Schmidt et al.[4]. The interaction-up genes include many ISGs, are enriched for antiviral defence and type I IFN signalling, and overlap with the canonical epithelial IAV response genes[17]. In wt cells, IFNB1 was not significantly induced at 6 hpi, which is in line with the block of IRF3 activation by NS1[1] and with the NS1-dependent targeting of IFN-related host mRNAs during IAV shutoff[18]. In infected 6KR cells, IFNB1 was higher than in infected wt cells. The interaction term for IFNB1 was not significant, however, so these data do not show that the induction itself differs between the genotypes; with few IFNB1 reads in wt cells, the wt induction is estimated imprecisely. The stronger ISG response of 6KR cells fits the model of Schmidt et al., in which ERV-derived dsRNA from derepressed loci adds to the viral stimulus[4]. The ERV transcripts themselves could not be examined here: they were quantified in the original study by a separate repeat-aware mapping, and the deposited count tables contain only annotated genes.

4.2 Genotype markers read out the TRIM28 repressor state

Ten of the 13 candidate genes are genotype markers, and five of them are KRAB-ZNF genes. KRAB-ZNF proteins recruit TRIM28 to repress transposable elements[2], and loss of TRIM28 in adult human cells derepresses both ERVs and ZNF genes[3]. The up-regulation of ZNF genes in 6KR cells, which Schmidt et al. also reported[4], is therefore the expected signature of impaired TRIM28 function. ADAMTS2, the candidate with the largest genotype effect, is also one of the two most prominently labelled genes of the mock comparison in the original study, and the cell-adhesion terms of this analysis match theirs. For ADAMTS2, TNNT1, ABI3BP, GLIPR1 and SULT2B1, no link to influenza or to TRIM28 was found in the literature; they are data-driven markers of the 6KR state in this cell line. **These genes mainly separate the genotypes, not infected from uninfected cells.** Whether they also report the infection-induced loss of TRIM28 SUMOylation in wt cells cannot be read from these data: infection changed several of them in wt cells, but in both directions (Table 2).

4.3 Infection markers and the host shutoff

The three infection markers need a different reading. IAV degrades most host mRNAs, but short transcripts and mRNAs of cell maintenance processes such as oxidative phosphorylation are refractory to this shutoff[19], and the RNAs that escape PA-X-mediated degradation are enriched for translation-related genes including ribosomal proteins[20]. DESeq2 normalisation assumes that most genes do not change; under a global loss of host mRNA, transcripts that resist the shutoff therefore appear induced. This most likely explains the dominance of ribosome and oxidative phosphorylation terms among the infection-induced genes and makes the induction of RPL30, and plausibly of the short housekeeping transcript OST4, a relative rather than an absolute increase. ARRB2 encodes β-arrestin 2, which promotes virus-induced IFN-β production through cGAS-STING in macrophages and is degraded by viruses as an immune-evasion strategy[21]; whether it has a similar role in IAV-infected epithelial cells is open. The classical ISG OASL, which enhances RIG-I signalling[22], was the most important gene of both final models but did not pass the stability rule, probably because many infection genes are about equally strong, so the infection half of the Model 2 selection changes between folds.

4.4 What changed compared with the course analysis

Two problems of the course analysis were fixed. First, the factors of the DESeq2 model had not been relevelled, so the alphabetical reference levels (6KR, infected) were used; the contrasts called “infection” and “genotype” therefore described the infection effect in 6KR cells and the genotype effect in infected cells. Second, the Random Forest was evaluated on repeated 60/40 splits, which left four test samples per split, so the accuracy could only take the values 0, 0.25, 0.5, 0.75 and 1 and changed with every random seed; in some runs, recursive feature elimination was applied to all samples before splitting. Nested LOOCV with the gene selection inside every fold avoids this selection bias[11,12].

4.5 Limitations

The study has 12 samples, three per group, from one cell line (A549, a lung carcinoma line), one virus strain, one high MOI and one time point, so the candidates describe this experimental system and not a patient population. The wt-infected libraries were the smallest, which increases the noise of weakly expressed genes in this group. The accuracy of Model 2 is likely optimistic, not only because of the small sample size, but also because its gene-selection rule was chosen after the Model 1 result on the same samples; only an independent dataset can confirm it. Each genotype is one lentivirally reconstituted population of the same knockout clone, so the genotype effect can also contain differences between the two transduced populations, for example in transgene expression, and not only the loss of the SUMOylation sites. Diagnostic host-response signatures of influenza are measured in blood and are dominated by ISGs[6–8]; the genes found here in an epithelial cell line would have to be tested in such cohorts before they could be called biomarkers for diagnosis.

4.6 Conclusion

Infection and the TRIM28 genotype can be separated from the transcriptome of these cells: the infection effect by a large margin, the genotype effect reliably only with a gene selection that accounts for the 2 × 2 design, a rule that still needs validation on independent data. The candidate genes mainly reflect TRIM28 repressor function and the host-shutoff pattern, while the amplified IFN

response of 6KR cells appears in the interaction analysis; both are consistent with the SUMO switch model of the original study.

Abbreviations

Gene symbols follow the HGNC nomenclature and are not listed.

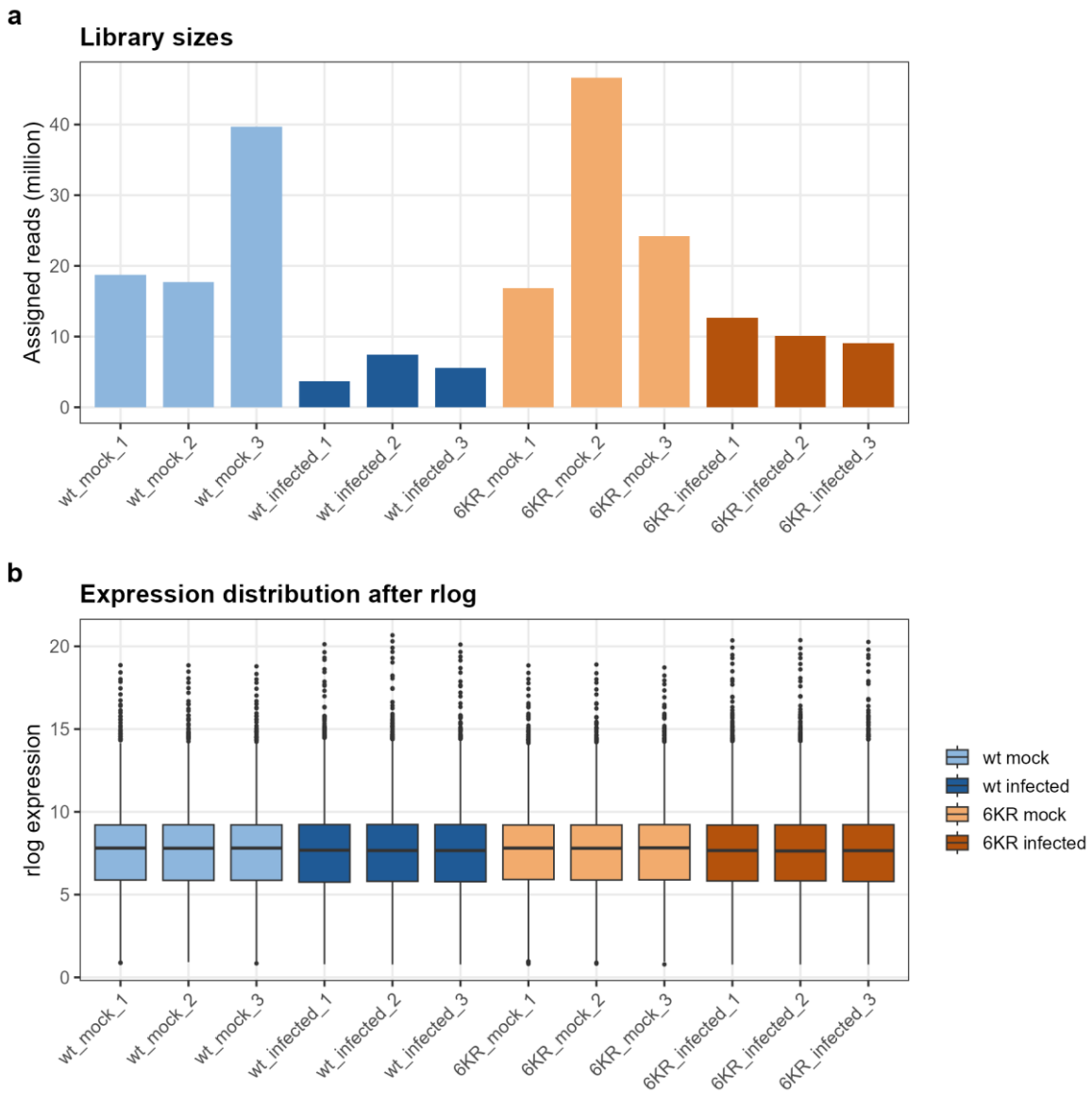
Abbreviation	Meaning
6KR	TRIM28 mutant in which the six SUMOylated lysines (K) are replaced by arginines (R)
BH	Benjamini-Hochberg (adjustment of p-values for multiple testing)
cGAS-STING	cyclic GMP-AMP synthase – stimulator of interferon genes (DNA-sensing pathway)
CI	confidence interval
DEG	differentially expressed gene
dsRNA	double-stranded RNA
ER	endoplasmic reticulum
ERV	endogenous retrovirus
FC	fold change
GEO	Gene Expression Omnibus
GO, GO:BP	Gene Ontology, Gene Ontology Biological Process
GRCh38	human reference genome, Genome Reference Consortium build 38
hpi	hours post infection
IAV	influenza A virus
IFN	interferon
IRF3	interferon regulatory factor 3
ISG	interferon-stimulated gene
KAP1	KRAB-associated protein 1 (other name of TRIM28)
KEGG	Kyoto Encyclopedia of Genes and Genomes
KRAB-ZNF, KRAB-ZFP	Krüppel-associated box zinc-finger protein
LOOCV	leave-one-out cross-validation
LRT	likelihood-ratio test (DESeq2 test of the four-group model against an intercept-only model)
MAVS	mitochondrial antiviral-signalling protein
MOI	multiplicity of infection
NS1	non-structural protein 1 of influenza A virus
PA-X	host-shutoff endoribonuclease of influenza A virus, encoded by the PA (polymerase acidic) gene
PCA	principal component analysis
REAC	Reactome pathway database (in figure labels)
RIG-I	retinoic acid-inducible gene I (cytosolic RNA sensor)
rlog	regularised logarithm (DESeq2 transformation)
RNA-seq	RNA sequencing
SUMO	small ubiquitin-like modifier
TRIM28	tripartite motif-containing protein 28
wt	wild type

References

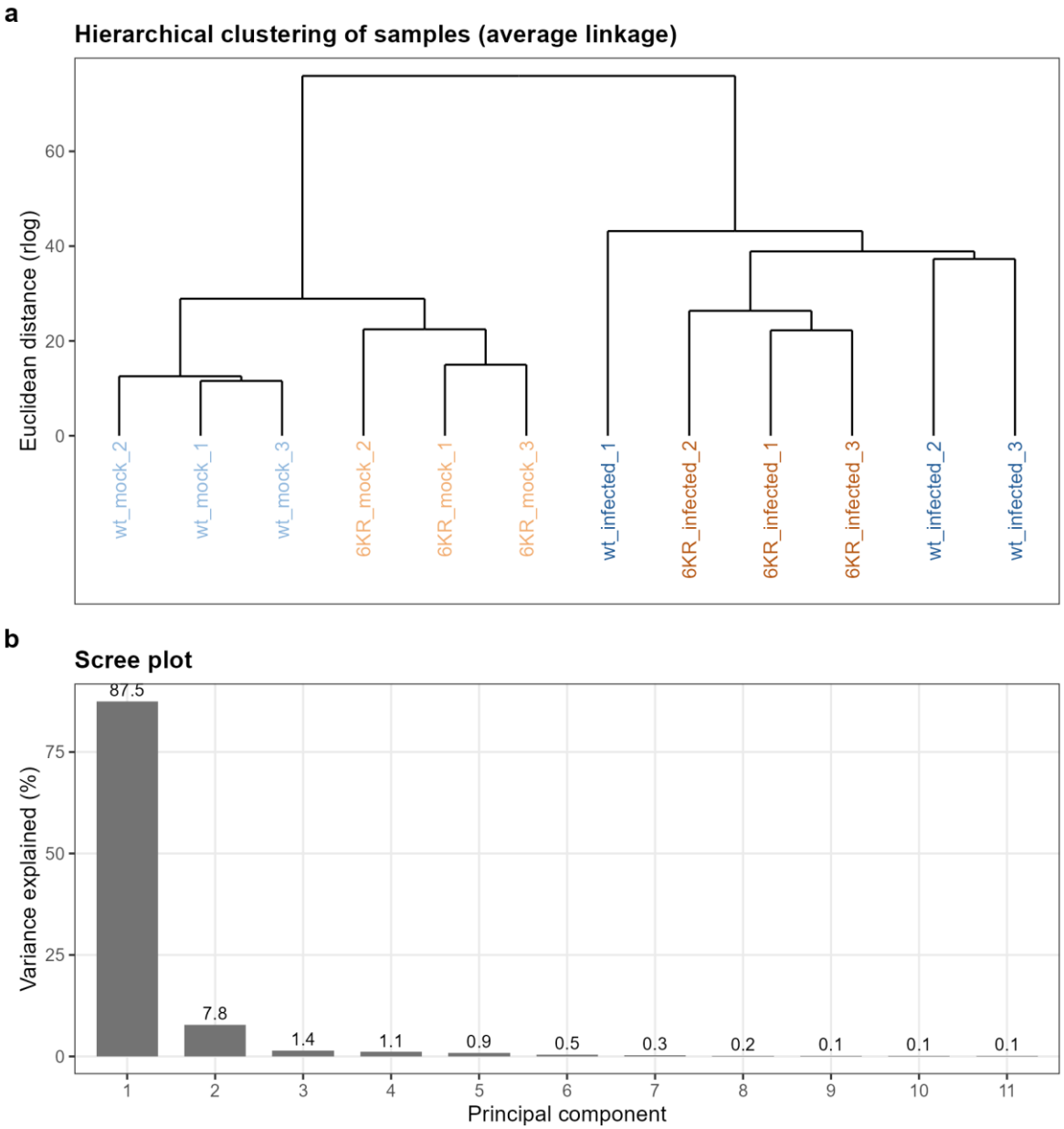
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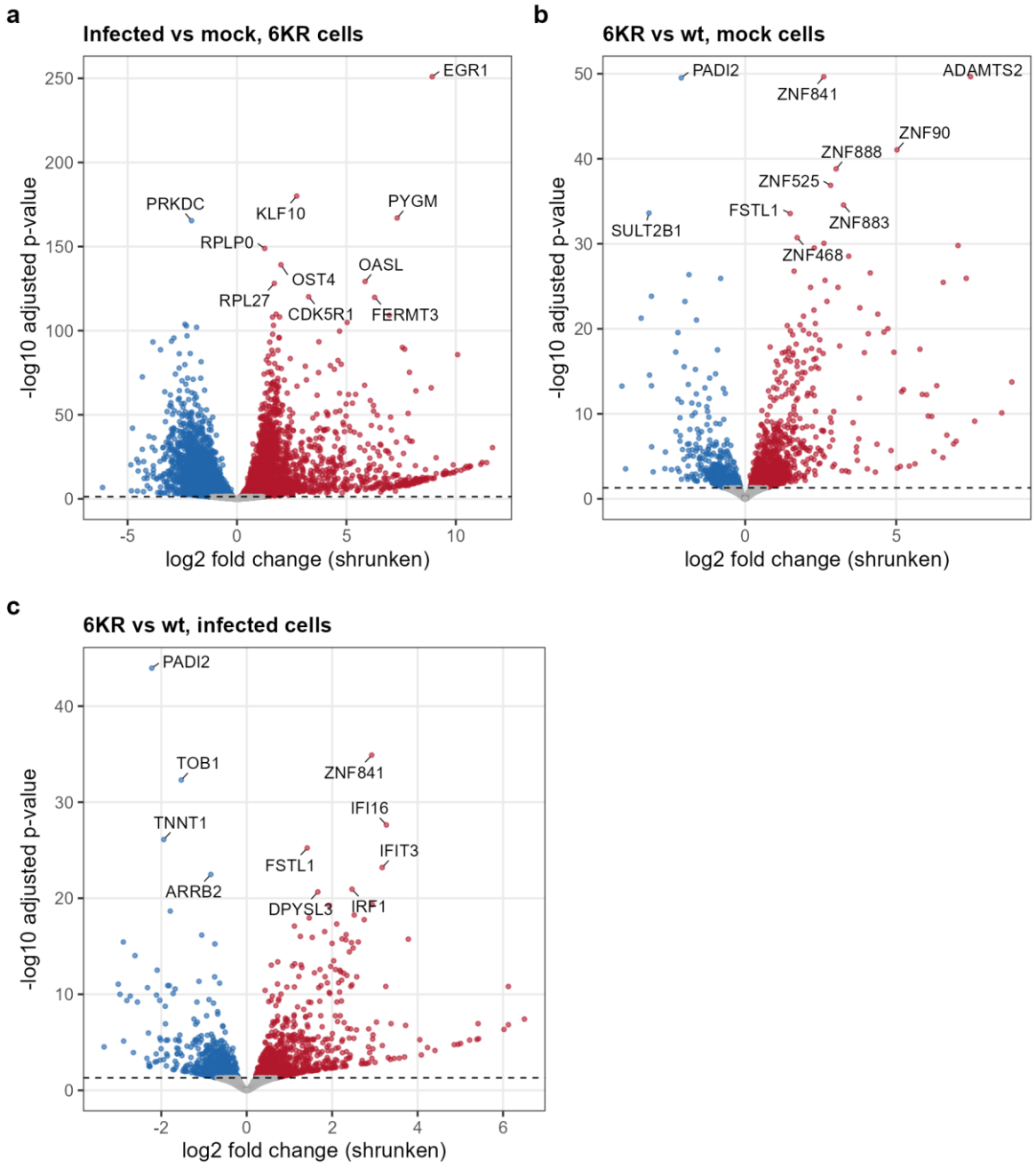
Supplementary figures



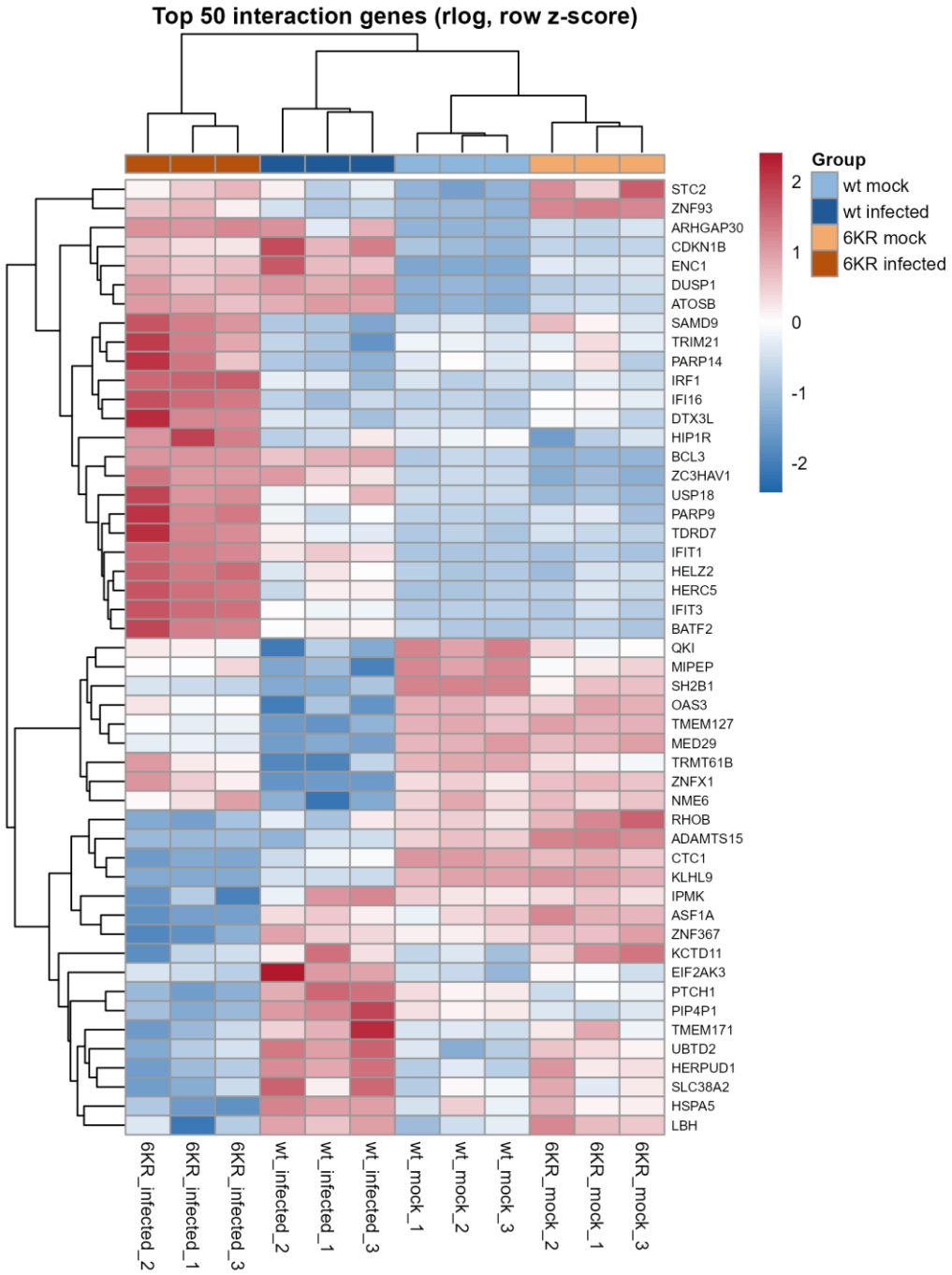
Supplementary Fig. 1| **Library sizes and expression distributions. a**, Assigned reads per sample. **b**, rlog expression per sample after filtering.



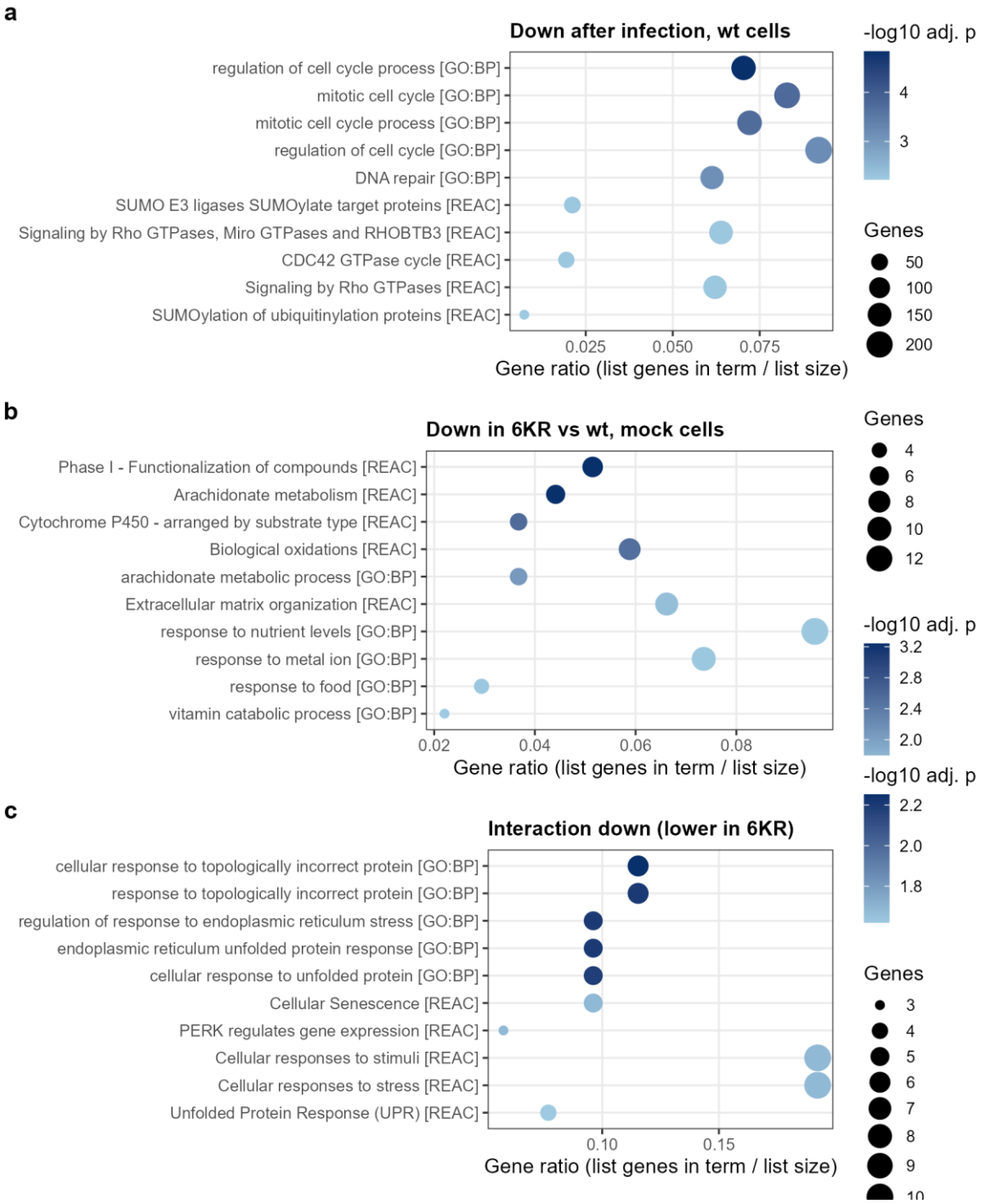
Supplementary Fig. 2| **Sample clustering. a**, Hierarchical clustering of samples on all filtered genes (Euclidean distance, average linkage). **b**, Variance explained by the principal components of the 500 most variable genes.



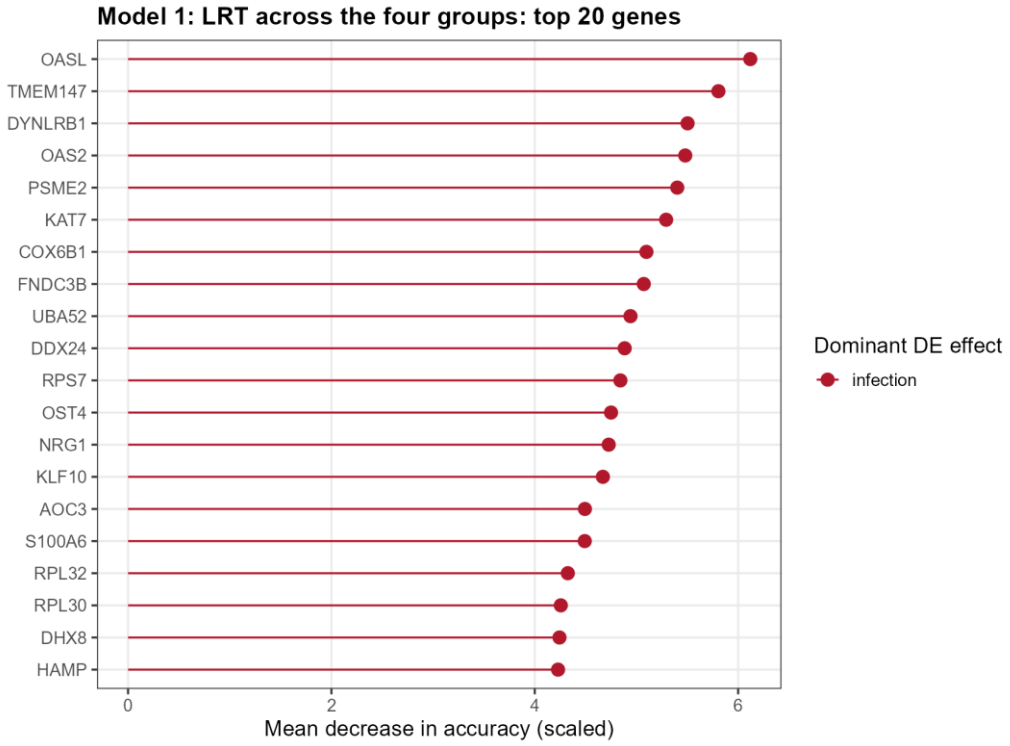
Supplementary Fig. 3| **Further contrasts.** Volcano plots for **a**, infection in 6KR cells; **b**, 6KR vs wt in mock cells; **c**, 6KR vs wt in infected cells.



Supplementary Fig. 4| **Interaction genes.** rlog expression of the 50 most significant interaction genes, scaled per gene (z-score).



Supplementary Fig. 5| **Over-representation analysis of the down-regulated lists. a**, Genes repressed by infection in wt cells. **b**, Genes down-regulated in mock-infected 6KR vs wt cells. **c**, Interaction-down genes (lower infection effect in 6KR than in wt cells).



Supplementary Fig. 6| **Model 1 importance.** The 20 most important genes of the final Model 1, coloured by their dominant effect in the differential expression analysis.